

Use Cases & End-to-End Journeys

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Document role. This is the Volume 1 (Product & Requirements) nano-detail spec enumerating HelixVPN's primary **end-to-end use cases**, each rendered as a step-by-step **journey** with a Mermaid sequence diagram, explicit preconditions and postconditions, and the exact **components and APIs** exercised. It turns the three roles, seven principles, and parity matrix of the Volume 1 overview (`00-product-scope-and-principles.md`) into runnable narratives that the §11.4.169 test types and the 8-criteria MVP DoD trace onto.

Document is a SPEC. It describes *what the user does and what the system does in response*; it does not build the product. Each journey is a candidate e2e / full-automation test scenario (§11.4.98 — fully self-driving, no manual step after start) and is written so its postconditions are captured-evidence assertions, not prose claims (§11.4 / §11.4.69).

Evidence base. Cross-references the Volume 1 overview (roles §3, two-way / multi-network §4, parity matrix §9, scope §10, decisions D1/D4/D6), the coordinator streaming contract (`v03-control-plane/svc-coordinator.md` — WatchNetworkMap, the < 1 s convergence + revoke SLOs, the reaction table), the telemetry presence + audit model (`v03-control-plane/svc-telemetry.md`), the no-logging invariant (`v05-security/no-logging-as-code.md`), and the transport escalation ladder (`v02-data-plane/transport-selection-ladder.md`). Research is cited by the overview's ids ([04_ARCH §N], [04_P1]). Anything unproven is marked UNVERIFIED per §11.4.6.

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1. Use-case catalogue & actor model

The actors are the three roles of the overview plus the human personas (00–... §3, §5).

Actor	Role / persona	App
End user	Mullvad-style privacy/lab user	Helix Access
Network operator	exposes a LAN safely	Helix Connector (daemon)
Admin	manages tenants/users/devices/policy	Helix Console
Gateway control plane	Go (identity/registry/ipam/pki/policy/coordinator/events/telemetry/api)	server
Gateway edge	Rust data plane (transports + WireGuard)	server

UC	Title	Primary actor	Phase	Parity / differentiator
UC-01	Consumer first connect (privacy exit)	End user	MVP	F1, F7, F14
UC-02	Switch exit / server	End user	MVP	F1, X1
UC-03	Enable multihop	End user	P2	F10
UC-04	Admin enrolls a device & sets policy	Admin	MVP	F15, F17, X1
UC-05	Connector advertises a route	Network operator	MVP	X1, X2
UC-06	Censorship-evade via transport escalation	End user	MVP	F2, F6, F7, F8
UC-07	Kill-switch trips on network drop	End user	MVP	F11, F13

Each journey below maps to the MVP Definition-of-Done acceptance criteria of the overview §10.2 where applicable, and each is a §11.4.98-compliant fully-automatable e2e scenario (no human step after start; non-introspectable UI falls back to the §11.4.117 pixel oracle).

2. UC-01 — Consumer first connect (privacy exit)

Goal. A new end user installs Access, enrolls anonymously (no email), and gets a full-tunnel privacy exit to the internet — the Mullvad use case.

Preconditions.

- A reachable Gateway (self-hosted or managed) with a public endpoint.
- An enrollment token (anonymous device-token path, no PII) OR an OIDC login.
- Access app installed on a supported platform (iOS/Android/Linux for MVP).

Postconditions (captured-evidence assertions).

- The device holds an overlay IP and a short-lived mTLS device cert.
- A WireGuard tunnel is established; the chosen exit carries the device's internet traffic (full-tunnel).
- A `presence:{tenant}:{device}` key exists in Redis (TTL'd); **no** durable connection row exists (NFR-300 / no-logging).
- An `device.enrolled` audit row exists (control action); **no** traffic audit.

Components / APIs exercised. identity (enroll), pki (device cert), registry (device + overlay-IP), ipam (overlay IP), coordinator (WatchNetworkMap), helix-core (tunnel + ladder), gateway edge (transport

- WG), telemetry (presence + audit).

sequenceDiagram

```
autonumber
participant U as Access app (helix-core)
participant ID as identity/pki
participant REG as registry/ipam
participant CO as coordinator
participant ED as edge (Rust+WG)
participant TEL as telemetry
U->>U: generate WG keypair on-device (private NEVER leaves, C6)
U->>ID: enroll(token | OIDC) + WG pubkey
ID->>ID: validate token; issue short-lived mTLS device cert
ID->>REG: register device; allocate overlay IP (ipam, ULA/48)
ID-->>U: device cert + overlay IP
U->>CO: WatchNetworkMap(mTLS, known_version=0)
CO-->>U: snapshot {self overlay IP, gateway endpoint, transport
order, peers per policy}
U->>ED: dial transport (ladder: plain-udp first) → WG handshake
ED-->>U: WG handshake complete → tunnel up
U->>ED: full-tunnel internet traffic via exit
U->>TEL: heartbeat → MarkOnline (presence TTL); device.enrolled
audited
Note over U,TEL: postcondition: tunnel up + presence key + NO
durable connection row
```

Honest boundary (§11.4.6). On iOS the tunnel runs inside the NE memory ceiling (NFR-500) — first-connect on iOS is gated by Phase-0 G3 and is UNVERIFIED until that on-device capture exists. The journey's logic is platform-neutral; the iOS resource fit is the open risk, not the flow.

3. UC-02 — Switch exit / server

Goal. A connected user picks a different exit (privacy server) or a different joined network, and traffic re-routes without dropping the app session.

Preconditions. UC-01 complete (device connected, has overlay IP + cert). The target exit/network is one the device's policy grants (need-to-know, C4).

Postconditions.

- The new exit/network carries traffic; the old one no longer does.
- Convergence to the new route is **< 1 s** (NFR-003) via a MapDelta, with **no** client restart (DoD-5 spirit).
- The kill-switch never opens a leak during the transition (NFR-404).

Components / APIs. Access UI (helix-ui ExitPicker), helix-core (re-pin/reconnect), coordinator (delta), edge (re-route), policy (VisibleTo / AllowedIPs).

sequenceDiagram

```
autonumber
participant U as Access app
participant C0 as coordinator
participant ED as edge
U->>U: user selects new exit/network in ExitPicker
U->>C0: (re)WatchNetworkMap / pin selection
C0->>C0: recompute self map (need-to-know filter, C4)
C0-->>U: MapDelta (upsert new peer/exit, remove old) version++
Note over C0,U: NFR-003 p99 < 1 s, no restart
U->>ED: reconcile WG peers to new exit
ED-->>U: traffic now via new exit; old peer removed
Note over U: kill-switch holds – no off-tunnel leak during switch (NFR-404)
```

Need-to-know enforcement (C4). The user can only switch to an exit/network their compiled policy grants — a non-granted target never appears in the snapshot (NFR-402), so the UI cannot offer it. This is the multi-network (X1) differentiator surfacing as a routing choice.

4. UC-03 — Enable multihop

Goal. A privacy-conscious user enables entry/exit separation (multihop) so the entry node never sees the exit destination and vice-versa (parity F10). Phase 2.

Preconditions. P2 multihop available; user connected (UC-01). The tenant's policy/topology offers ≥ 2 hops in distinct jurisdictions.

Postconditions.

- Traffic traverses nested WireGuard (entry $\rightarrow$ exit), control-plane orchestrated.
- No single hop holds both the user identity and the destination.
- Latency rises (extra hop) within the obfuscation-aware budget (NFR-007) — exact added latency UNVERIFIED (P2 benchmark owns it).

Components / APIs. Access UI (multihop toggle), helix-core (nested WG), coordinator (per-hop keys in the map), policy (hop authorization), edge.

```
sequenceDiagram
    autonumber
    participant U as Access app
    participant C0 as coordinator
    participant E1 as entry hop
    participant E2 as exit hop
    U->>U: enable multihop (entry, exit) toggle
    U->>C0: WatchNetworkMap (multihop request in policy)
    C0-->>U: map with per-hop keys (entry pubkey, exit pubkey) [P2]
    U->>E1: nested WG: outer tunnel to entry
    E1->>E2: inner tunnel entry->exit (entry cannot read inner dest)
    E2-->>U: internet via exit
    Note over E1,E2: entry sees user, not dest; exit sees dest, not user (F10)
```

Phase boundary (§11.4.6). Multihop is explicitly OUT of MVP scope (overview §10.2 → P2). This journey is the P2 contract; its postconditions are UNVERIFIED until the P2 nested-WG implementation and its e2e capture exist.

5. UC-04 — Business admin enrolls a device & sets policy

Goal. An admin (Console) creates a user/device enrollment, then writes a default-deny ACL granting a group access to a specific network/port — reflected on the edge in < 1 s with no restart.

Preconditions. Admin authenticated to Console (RBAC admin/operator). A tenant exists; at least one connector has advertised a network (UC-05).

Postconditions.

- An enrollment token is minted (audited enroll_token.mint); the device enrolls (UC-01) and appears in the registry.
- A policy edit (group:contractors → net:warehouse:554/tcp) compiles to per-peer AllowedIPs + an edge verdict map and reaches affected agents in < 1 s (NFR-003 / DoD-5).
- Audit holds policy.create + policy.activate control rows; **no** traffic.

Components / APIs. Console (helix-ui admin + REST /v1/...), api (Gin REST + WS/SSE), identity (enroll token), policy (compiler), coordinator (fan-out), telemetry (audit).

```
sequenceDiagram
    autonumber
    participant AD as Console (admin)
    participant API as api (Gin REST)
    participant POL as policy compiler
    participant BUS as events bus
```

```

participant C0 as coordinator
participant ED as edge
AD-->>API: POST /v1/enroll-tokens (mint) → audited
enroll_token.mint
API-->>AD: token (shown once; never logged §11.4.10)
AD-->>API: PUT /v1/policy (group:contractors → net:warehouse:554/
tcp)
API-->>POL: compile (default-deny, fail-closed)
POL-->>BUS: events:policy {policy.compiled, version}
BUS-->>C0: deliver
C0-->>C0: projectionDiff → minimal affected set
C0-->>ED: MapDelta (AllowedIPs + verdict map) version++
Note over C0,ED: NFR-003 p99 < 1 s, no restart (DoD-5)
AD-->>API: GET /v1/audit → policy.create + policy.activate rows
(control-only)

```

Authorization boundary. GET /v1/audit requires admin/operator; member is denied; RLS is the floor so even a mis-scoped query returns only the caller's tenant rows (NFR-408). The minted token is shown once and never logged (§11.4.10).

6. UC-05 — Connector advertises a route

Goal. A network operator installs the Connector daemon inside a private LAN; it dials **outbound** to the Gateway and advertises the LAN's CIDRs — no inbound port-forward, ever (P3 / X2).

Preconditions. Connector daemon installed on a host inside the LAN. An enrollment token for the connector. Outbound reachability to the Gateway.

Postconditions.

- The connector holds a device cert + overlay identity and an attached site (4via6 site id, D4).
- Its advertised CIDRs are registered; authorized users' maps gain reachable peers/routes for that network.
- Overlapping RFC1918 ranges across connectors are disambiguated by the overlay addressing scheme (ULA /48 + 4via6 / per-network NAT fallback, D4).
- **No** inbound port was opened on the LAN's router (P3 assertion).

Components / APIs. Connector (helix-core advertise/route mode), identity / pki, registry (route.advertised), ipam (site), coordinator (reaction connector.attached / connector.prefixes.changed), policy.

```

sequenceDiagram
    autonumber
    participant CN as Connector daemon (in LAN)
    participant ID as identity/pki
    participant REG as registry/ipam
    participant C0 as coordinator
    participant U as authorized Access user
    CN-->>ID: outbound dial + enroll(token) + WG pubkey

```

```

ID-->>CN: device cert; overlay identity; site id (4via6, D4)
CN-->>REG: advertise CIDRs (route.advertised)
REG-->>C0: events {connector.attached,
connector.prefixes.changed}
C0-->>C0: recompute maps for nodes whose policy grants these
CIDRs
C0-->>U: MapDelta (new reachable network via this connector)
Note over CN: NO inbound port opened on the LAN router (P3 / X2)
Note over C0,U: overlapping 192.168.x ranges disambiguated by D4
overlay addressing

```

Conflict handling (§11.4.6). Two connectors advertising the same RFC1918 range raise `route.conflict.detected`; the coordinator does **not** push a delta for it — the conflict surfaces to Console and 4via6 site disambiguation resolves it (D4). The concrete addressing scheme is frozen in the Phase-1 IPAM design; until then the collision resolution detail is UNVERIFIED.

7. UC-06 — Censorship-evade via transport escalation

Goal. A user on a censored network where plain WireGuard/UDP is blocked automatically escalates the transport ladder until an obfuscated rung (MASQUE/QUIC) carries the WG handshake — Mullvad's "automatic obfuscation" (F7), looking like ordinary web traffic.

Preconditions. Device enrolled (UC-01). The current network blocks plain-UDP WG (DPI / UDP block). The ladder order is the default or a coordinator-pushed regional prior.

Postconditions.

- The ladder advances monotonically within the attempt (L-I8) until a rung completes a **WG handshake** (carrier-connect alone is not success, I2).
- The working rung is remembered per-network (fingerprint hash + rung index, **no** SSID plaintext, **no** endpoint, no per-connection log — I5).
- Aggregate escalation telemetry is recorded (counts only, no per-user flow).
- Throughput on the obfuscated rung meets the DPI-survival bar ($\text{NFR-002} \geq 50\%$ of plain — TARGET UNVERIFIED until G2).

Components / APIs. `helix-core ladder (ladder.rs)`, `helix-transport (plain-udp / lwo / masque-h3 / ...)`, `edge (transport termination)`, `coordinator (TransportPolicy.order regional prior)`, `telemetry (aggregate ladder outcomes)`.

flowchart TD

```

Start([connect on censored network]) --> Seed{seed cursor:<br/>per-net memory -><br/>regional prior -> default}
Seed --> D0[dial plain-udp]
D0 --> H0{WG handshake<br/>within budget?}
H0 -- no, UDP blocked --> E1[escalate: cursor+1 · emit
Reconnecting]
E1 --> D1[dial lwo]

```

```

D1 --> H1{WG handshake?}
H1 -- no --> E2[escalate cursor+1]
E2 --> D2[dial masque-h3 / CONNECT-UDP HTTP-3]
D2 --> H2{WG handshake?}
H2 -- yes --> R[remember fingerprint→rung · Connected]
H0 -- yes --> R
H1 -- yes --> R
R --> T[aggregate ladder telemetry · counts only · no per-user
flow I5]

```

No-logging on the evade path (§11.4.6). Per-network memory persists only a fingerprint *hash* + last-good rung index — never the SSID, never the server endpoint, never a timestamped per-connection record (I5). The escalation event trace IS the captured evidence (L-I9), and it carries no destination. The $\geq 50\%$ DPI-survival throughput is a TARGET pending Phase-0 G2.

8. UC-07 — Kill-switch trips on network drop

Goal. The user's underlying network drops (or the tunnel fails); the kill-switch blocks all off-tunnel traffic so nothing leaks, and DNS stays forced through the tunnel (F11 + F13).

Preconditions. Device connected with kill-switch enabled (default-on for the privacy posture). The OS firewall integration is active (*helix-core* state machine + per-OS firewall driver).

Postconditions.

- On tunnel-down, **0** packets traverse the physical interface off-tunnel (NFR-404 leak test).
- **0** plaintext DNS queries escape off-tunnel (NFR-405).
- On reconnect (ladder re-runs, UC-06), the kill-switch lifts only once a tunnel is re-established — never a leak-window during the gap.

Components / APIs. *helix-core* (kill-switch + DNS state machine), per-OS firewall shim (*nftables*/*WFP*/*NE* rules), coordinator (reconnect map), edge.

```

stateDiagram-v2
    [*] --> Connected
    Connected --> TunnelDown: network drop / handshake loss
    TunnelDown --> Locked: kill-switch engages (firewall blocks off-
tunnel + plaintext :53)
    Locked --> Locked: any app traffic dropped (0 leak, NFR-404/405)
    Locked --> Reconnecting: ladder re-runs (UC-06)
    Reconnecting --> Connected: WG handshake re-established → kill-
switch lifts
    note right of Locked
        No off-tunnel packet and no plaintext DNS escapes
        during the gap – the leak window is closed by construction.
    end note

```


Verification (§11.4.69). The kill-switch postconditions are proven by a packet-capture leak test on the physical interface during a forced tunnel-drop: a non-zero off-tunnel packet count is a FAIL. The capture IS the evidence; "no leak observed" without the capture is a PASS-bluff (§11.4).

9. Cross-cutting postcondition invariants

Every journey above, regardless of actor, must leave the system in a state that satisfies these invariants — they are the union of the seven principles and the no-logging guarantee surfacing as per-journey postconditions:

Invariant	Holds in every journey because	Verified by
Fail-static (P1)	the edge forwards from a cached verdict map; a control-plane blip during any journey never drops an established tunnel	chaos (kill control plane mid-journey)
Outbound-only (P3)	every actor <i>dials out</i> ; the gateway is the only public listener	security (no-inbound assertion)
Need-to-know (C4)	every map a journey delivers is policy-filtered; a non-granted peer never appears	meta-test (skip-filter mutation FAILs)
< 1 s convergence (P4)	every map change in a journey is a minimal MapDelta	performance (helix_reconcile_seconds)
No-logging (P7)	no journey writes a durable connection/traffic row; presence is ephemeral; audit is control-only	security (schema-lint, deployed DB)
Anonymous identity (F15)	first-connect (UC-01) needs no email/SSO	integration (anonymous enroll)

Anti-bluff coupling (§11.4 / §11.4.98). Each journey is authored as a fully self-driving e2e scenario: no human action after start, deterministic across re-runs (§11.4.50), captured evidence per postcondition. A journey that "passes" without its postcondition captures (presence key present, no durable row, delta on wire, zero leak) is a PASS-bluff at the use-case layer.

10. Use-case → component → API → test traceability

UC	Components	Key APIs / streams	DoD / NFR	§11.4.169 test type
UC-01	identity, pki, registry, ipam, coordinator, core, edge, telemetry	enroll; WatchNetworkMap; transport dial	DoD-1/2, NFR-307/304	e2e / full-automation
UC-02	core, coordinator, edge, policy	WatchNetworkMap delta	DoD-5, NFR-003/402	e2e
UC-03	core, coordinator, edge, policy	map w/ per-hop keys (P2)	F10, NFR-007	e2e (P2)
UC-04	api, identity, policy, coordinator, telemetry	REST /v1/...; events:policy; audit	DoD-3/5, NFR-003/408	e2e + integration
UC-05	core, identity, pki, registry, ipam, coordinator, policy	route.advertised; connector.attached	X1/X2, NFR-103/104	e2e + integration
UC-06	core (ladder), transport, edge, coordinator, telemetry	ladder; TransportPolicy.order	DoD-4, NFR-002/410/006	e2e (DPI-sim)
UC-07	core (kill-switch/ DNS), firewall shim, coordinator, edge	firewall state machine	DoD-7, NFR-404/405	security / e2e (leak test)

flowchart LR

subgraph Actors

EU[End user]

OP[Network operator]

AD[Admin]

end

EU --> UC01[UC-01 connect] --> CORE[helix-core + edge]

EU --> UC02[UC-02 switch] --> COORD[coordinator]

EU --> UC06[UC-06 evade] --> LAD[ladder + transport]

EU --> UC07[UC-07 kill-switch] --> FW[firewall shim]

OP --> UC05[UC-05 advertise] --> REG[registry/ipam]
AD --> UC04[UC-04 enroll+policy] --> POL[policy + api]
COORD --> SLO[< 1 s convergence SLO]
LAD --> NL[no-logging: counts only]
FW --> LEAK[zero-leak capture]

Phase honesty (§11.4.6). UC-01/02/04/05/06/07 are MVP-scoped; UC-03 (multihop) is P2. P2/P3 postconditions are UNVERIFIED until their phase's implementation and captured e2e evidence exist — they are the forward contract, never asserted as already working.

Sources verified

- docs/research/mvp/final/00-product-scope-and-principles.md — three roles + control relationships (§3), two-way / multi-network exact meaning (§4), personas + app classes (§5), Mullvad-parity matrix F1–F17 + X1–X5 (§9), phase scope + 8-criteria MVP DoD (§10), decisions D1 (MASQUE)/D4 (ULA+4via6)/D6 (asymmetric per-leg) (§11).
- docs/research/mvp/final/v03-control-plane/svc-coordinator.md — the WatchNetworkMap streaming handler, known_version resume, the reaction table (device.enrolled/device.revoked/connector.attached/connector.prefixes.changed/policy.compiled), need-to-know VisibleTo filtering (C4), the < 1 s convergence + < 1 s revoke SLOs (§7).
- docs/research/mvp/final/v03-control-plane/svc-telemetry.md — ephemeral Redis presence (MarkOnline/Refresh/Offline, TTL), the control-action audit vocabulary
 - meta-shape guard, counts-only metrics.
- docs/research/mvp/final/v05-security/no-logging-as-code.md — the no-logging postcondition invariants (no durable connection row, presence ephemeral, audit control-only) every journey must satisfy.
- docs/research/mvp/final/v02-data-plane/transport-selection-ladder.md — the escalation ladder behind UC-06 (seed order, monotonic cursor L-I8, WG-handshake success criterion I2, per-network memory + no-logging I5, aggregate telemetry).

Constitution bindings applied: §11.4.44 (revision header), §11.4.6 (no-guessing — P2/P3 postconditions + unproven targets marked UNVERIFIED), §11.4.98 (every journey authored as a fully self-driving e2e scenario, no manual step after start), §11.4.50 (deterministic re-run), §11.4.69/§11.4.107 (each postcondition is a captured-evidence assertion), §11.4.117 (pixel-oracle fallback for non-introspectable UIs), §11.4.25 (use-case → component → test coverage ledger §10), §11.4.65/§11.4.153 (HTML+PDF[+DOCX] exports follow in refinement).